

News Articles Classification Problem

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Abstract—In this report, we present an approach to the news article classification problem. The method involves preprocessing the dataset, with particular attention to the textual content, to reduce noise and extract meaningful features, including also related numerical information. Two classifiers are employed, and their predictions outperform a naive baseline, achieving promising results for this challenging task.

I. PROBLEM OVERVIEW

The proposed competition is a classification problem focused on the categorization of online news articles. The goal of this competition is to correctly identify the thematic category of a news article based on its textual content and associated metadata.

The dataset provided for this task consists of approximately 80,000 instances, where each record represents a news article. For each article, the following information is available:

- **Metadata:** a unique identifier, the source (publisher), the publication timestamp, and the page rank of the source.
- **Textual content:** the title and the full body of the article.

The problem is a multi-class classification task with 7 predefined categories: International News (0), Business (1), Technology (2), Entertainment (3), Sports (4), General News (5), and Health (6).

The dataset used for training our model is not well balanced regarding the distribution of labels, instead has a predominance of records with label equal to 0, and few records labelled with 6, as showed in the graph in Figure 1.

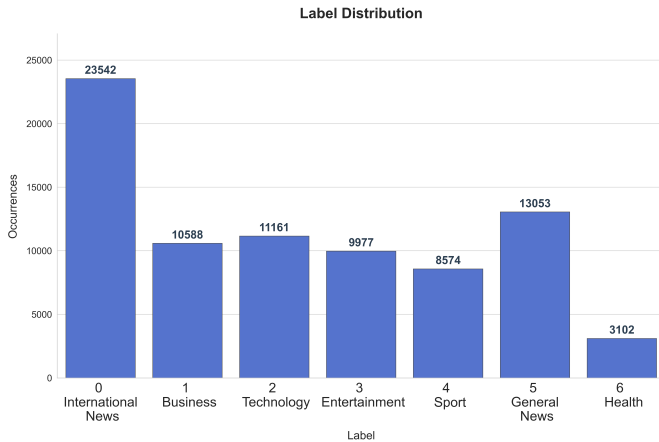


Fig. 1. Label distribution in the training dataset.

The dataset is divided into a *development set*, which includes the labels for training and validating the models, and an *evaluation set*, for which predictions must be submitted. The performance of the proposed solutions is evaluated using the *Macro F1-score*, a metric that ensures equal importance is given to all classes, regardless of their frequency in the dataset.

II. PROPOSED APPROACH

A. Dataset cleaning and preprocessing

The preprocessing phase aimed to standardize the textual content, remove noise, and enrich the dataset with informative features useful for the classification. The main information in the given dataset are included in the textual features (*title*, *article*, *source*). Since these data contained noise due to html, tag, incorrect conversion of characters from online, we applied a cleaning procedure to uniform the text.

The cleaning text procedure can be summarized in the following steps:

1) Title cleaning and source extraction

News titles were first cleaned by removing HTML artifacts, malformed encodings, and excessive whitespace. When present, source information appended at the end of the title was identified by the pattern source within parentheses, and extracted into a separate feature. This extracted source was then compared with the original source field provided in the dataset, and a binary variable *source_changed* was introduced to capture discrepancies between the two. This feature highlights cases in which news articles are republished by outlets different from the original publisher. Additionally, non-informative prefixes such as "editorial", "corrected", and "update" were removed from titles, as they do not convey semantic information relevant to the news content.

2) Article preprocessing and URL-based information extraction

The article body was cleaned by removing HTML tags, URLs, and noisy characters. Beyond basic cleaning, additional information was extracted directly from embedded URLs, including image alternative texts, domain names, and path-level keywords. These elements were transformed into contextual tokens and grouped into coarse-grained semantic categories. An exploratory analysis revealed a strong correlation between these URL-

derived features and the target labels, motivating their inclusion as auxiliary textual signals.

3) Text aggregation

A unified textual representation (*full_text*) was constructed by concatenating the cleaned title and the processed article content. This combined field serves as the primary input for text vectorization, ensuring that both headline-level and body-level information contribute to the classification process.

4) Source aggregation

Source names were normalized to address inconsistencies such as domain suffixes (e.g., .com, .net) and abbreviations. Frequently occurring sources with different notations were manually consolidated to reduce redundancy and improve the reliability of categorical features.

5) Duplicate handling and label disambiguation

Duplicate news items were identified based on identical title-article pairs. To avoid introducing noise into the learning process, duplicates associated with conflicting labels were resolved using a majority-voting strategy: for each duplicated item, the label occurring most frequently was retained, while ties were discarded. This approach preserves informative samples while preventing label ambiguity, improving upon a naive duplicate removal strategy used by default.

6) Rare source grouping

To mitigate feature sparsity and reduce dimensionality after one-hot encoding, sources with very low frequency were grouped under a common 'Other' source name. This prevents the model from overfitting to rare publishers while maintaining the most informative source-level distinctions.

To retrieve more information related to the textual part, we added some features in the dataset related to the *full_text*:

- Length of the article in terms of words;
- Ratio between title length and article length, which provides a measure of how informative or concise a headline is relative to the body of the article;
- Digit density, to capture stylistic patterns related to numerical content, which may differ across news categories and sources.

From the feature *timestamp* present in the dataset, we extracted other features:

- Hour of publication, encoded using a cyclic representation (sine and cosine transformation) to preserve the circular nature of time and avoid artificial discontinuities between adjacent hours (e.g., 23:00 and 00:00);
- Publication period of the day, by discretizing hours into coarse day periods (morning, afternoon, evening, and night);
- Ratio between title length and article length, which provides a measure of how informative or concise a headline is relative to the body of the article;

- A binary indicator was also introduced to distinguish weekday from weekend publications.

These features can be used to improve the classification.

Regarding the numerical features, we used the page rank. We noticed that the page rank was set equal to 5 for the majority of the record, but we noticed that records with a lower score only belonged to label 1, 2, 3, 4, in particular to label 2, and this information could help during the classification.

After text cleaning procedure, we used heterogeneous strategies for different type of data.

Textual features were modeled using *Term frequency inverse document frequency (tf-idf)*. In addition to the standard English stop-word list, a custom set of domain-specific stop words was defined to remove frequent but weakly informative terms commonly found in news articles (e.g., reporting verbs, temporal references, and weekdays). For word-level representations, both unigrams and bigrams were considered, with a maximum vocabulary size of 50,000 features and sublinear term-frequency scaling to reduce the influence of very frequent terms.

Separate Tf-idf vectorizers were applied to the main article text (*full_text*), image alternative text (*image*), and extracted keyword features (*keywords*) in order to preserve their distinct semantic contributions.

Categorical variables (*source* and publication time period *day_period*) were encoded using one-hot encoding, while numerical features were standardized using z-score normalization.

B. Model selection

Regarding model selection, several classification algorithms were initially evaluated to assess their performance on the available dataset.

Based on previous studies on news article classification ([1], [2], [3]), the following algorithms were tested:

- *RandomForestClassifier*: This algorithm leverages multiple decision trees trained on different subsets of the data and features to make predictions. By combining many trees, it typically avoids overfitting of a single decision tree while maintaining a degree of interpretability.
- *LinearSVC*: Linear Support Vector Classification is a margin-based classifier that uses a Support Vector Machine with a linear kernel [4]. It is particularly effective for text classification, as it handles high-dimensional sparse features efficiently and is robust to noise and outliers.
- *LogisticRegression (LR)*: Logistic Regression is a linear probabilistic model that estimates class probabilities based on the input features and performs classification by selecting the class with the highest probability.

The RandomForestClassifier exhibited inferior performance compared to the other two approaches; therefore, only LinearSVC and LogisticRegression were retained for the subsequent analysis, while RandomForestClassifier was used as a baseline for comparative purposes in the results section. For both selected classifiers, the optimal hyperparameter configurations were identified through a grid search procedure, as described in the following section.

C. Hyperparameters tuning

There are three main sets of hyperparameters to be tuned, for Tf-idf, LogisticRegression e LinearSVC defined in Table I. Hyperparameter tuning was performed using a grid-based search combined with a 3-fold cross-validation strategy.

TABLE I
HYPERPARAMETERS CONSIDERED

Model	Parameter	Values
Tf-idf	<i>ngram_range</i>	{(1,2),(1,3)}
	<i>min_df</i>	{1,2,3}
LogisticRegression	<i>C</i>	{1, 1.05, 1.1, 1.15, 1.2}
	<i>solver</i>	{'lbfgs'}
	<i>class_weight</i>	{'balanced'}
	<i>penalty</i>	{'l1','l2'}
LinearSVC	<i>C</i>	{0.1, 0.5, 0.85, 0.90, 0.95, 1}
	<i>penalty</i>	{'l1','l2'}
	<i>class_weight</i>	{'balanced'}
	<i>max_iter</i>	{1000,2000}

The hyperparameters reported in Table I were selected based on common practices in text classification and preliminary experiments.

- Tf-idf: The *ngram_range* controls which n-grams are included, allowing the model to capture contextual word sequences. Higher-order n-grams ((1,2) or (1,3)) are particularly useful for news classification. *min_df* removes very rare terms, reducing noise and dimensionality. Both parameters strongly affect feature quality.
- Logistic Regression: The key parameter is *C*, which regulates regularization and prevents overfitting. *penalty* allows l1 or l2 regularization, affecting sparsity and coefficient stability. *class_weight*=‘balanced’ addresses class imbalance, while the *solver*=‘lbfgs’ ensures efficient and stable optimization.
- LinearSVC: *C* and *penalty* are the most important parameters, controlling regularization and sparsity. *class_weight*=‘balanced’ handles class imbalance, and *max_iter* ensures convergence in high-dimensional sparse spaces.

Overall, the parameter ranges were defined to balance underfitting and overfitting, manage feature sparsity, and handle class imbalance, key factors in effective text classification.

III. RESULTS

The best configuration for Tf-idf was found for {*ngram_range* = (1,2), *min_df* = 2}, instead regarding the classification models we obtained this configurations:

- LogisticRegression: {*C*=1.15, *class_weight*=‘balanced’, *solver*=‘lbfgs’, *penalty*=‘l2’, *random_state*=42, *max_iter*=1000};
- LinearSVC: {*C*=0.09, *penalty*=‘l2’, *max_iter*=1000, *class_weight*=‘balanced’, *random_seed*=42, *dual*=False}.

Both classifiers achieved satisfactory and comparable performance. LinearSVC slightly outperformed Logistic Regression on this dataset. Previous studies on news classification ([2], [3]) also indicate that LinearSVC often performs better than Logistic Regression for high-dimensional text data.

The best-performing LinearSVC and LogisticRegression models were trained on all available development data and subsequently used to label the evaluation set. The overall results are reported in Table II.

TABLE II
PERFORMANCE COMPARISON

Model	Train-Test Score	Leaderboard Score
RandomForestClassifier	0.6236	0.721
LogisticRegression	0.7241	0.721
LinearSVC	0.7292	0.728

Compared to the RandomForestClassifier baseline (F1-macro = 0.6236), both linear models achieved higher F1 scores, confirming that LinearSVC and Logistic Regression are better suited for high-dimensional sparse features typical of text data. While the baseline provides a useful reference, its performance is limited because tree-based methods are less effective in exploiting sparse feature representations. The ensemble VotingClassifier further improved results by combining the predictions of both linear models, allowing the system to capture complementary information and reduce misclassifications, particularly in classes with overlapping textual content.

Analysing the confusion matrix (in Figure 2), we can observe that the most significant classification challenge is the semantic convergence between General News (Label 5) and International News (Label 0). The confusion matrix reveals a high degree of bidirectional error. This is primarily attributed to “thematic ambiguity”: analysis of top-frequency terms shows that both categories rely heavily on a shared political lexicon, including keywords such as “president,” “government,” “Iraq,” and “minister”. Consequently, the distinction between these classes often depends more on the editorial source’s categorization rather than unique linguistic markers, posing an inherent limit on the discriminative power of text-based models for these specific labels.

The “Entertainment” class (Label 3) is also a complex category in the dataset, characterized by significant dispersion across the confusion matrix. Although the model successfully identified many instances, this category exhibits substantial leakage into “International News” and “General News”, which account for 229 and 210 misclassifications, respectively.

In contrast to the news-centric labels, "Sports" (Label 4) and "Technology" (Label 2) demonstrate high classification accuracy with minimal leakage. These categories benefit from a "domain-specific nomenclature" that provides a strong signal for the classifying models.

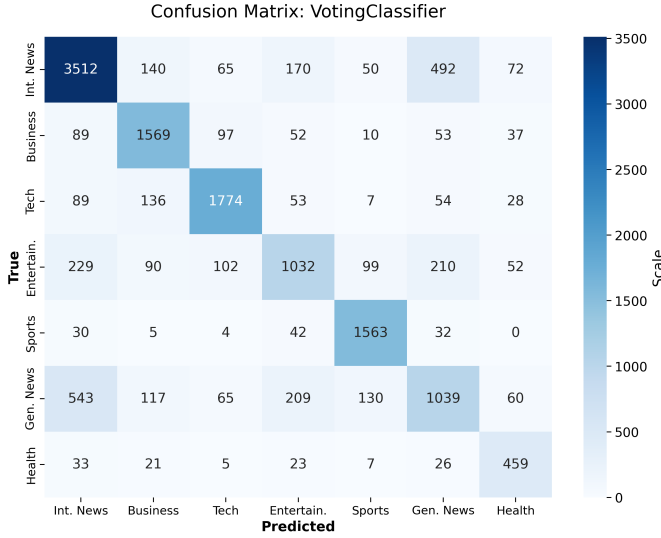


Fig. 2. Confusion Matrix obtained with VotingClassifier

IV. DISCUSSION

The experimental results demonstrate that the proposed approach provides a solid foundation for news classification.

However, several aspects could be explored to further improve performance:

- A more exhaustive hyperparameter search could be performed using grid search strategies. In the current setup, only a limited subset of parameters was explored, while a broader search over regularization strengths, n-gram configurations, and model-specific parameters could further improve performance and robustness.
- Additional cleaning steps for the text that may help reduce redundancy and improve semantic consistency in the textual representation. These include lemmatization or stemming, improved handling of named entities, and more refined noise removal for text.
- More sophisticated ensemble strategies, such as weighted voting schemes, could be investigated to better exploit the complementary strengths of different classifiers and further improve performance ([5]).
- The use of transformer-based architectures, such as BERT and its variants, could be explored. These models provide contextualized word representations and have achieved state-of-the-art performance in many text classification tasks. Prior work also suggests that MLP classifiers may offer competitive performance in similar scenarios [3].

It is worth noting that the obtained F1-macro of 0.73 represents a solid performance for this type of news

classification task, which involves multiple, partially overlapping categories. Some of the lower-performing labels, such as "General News", likely contain heterogeneous content that could plausibly belong to other categories, making accurate classification inherently difficult. Moreover, correlations among certain categories—for instance, "Health" and "Sport", or "Sport" and "Entertainment"—can introduce ambiguity in the textual patterns, further complicating the classification process. These observations suggest that future work could focus on more fine-grained category definitions or hierarchical classification strategies to mitigate label overlap and improve performance.

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